Mini Report: Unsupervised Refinement on Qwen3 Audio Embeddings

NXSCough Experiments

March 11, 2026

Pipeline Summary

1. Extract embeddings with `Qwen3_Extractor` (mean+std pooled representation).
2. Baseline supervised evaluation across 5 classifiers.
3. Harmonize embedding space with empirical Bayes ComBat.
4. Detect/remove outliers (Isolation Forest + centroid distance ensemble, 5%).
5. Estimate acoustic TB likelihood (kNN + GMM posterior ensemble).
6. Relabel TB samples into:
   - **Acoustic-TB** (high score),
   - **Label-only-TB** (low score), then filter label-only TB.

Qwen3 Extractor Architecture

- Backbone: audio tower of `Qwen3-Omni-30B-A3B-Thinking`, used as a **frozen feature extractor**.
- Audio pipeline:

  wav $\rightarrow$ Mel frontend $\rightarrow$ Qwen3 audio encoder ($3\times$ CNN stride) $\rightarrow$ frame embeddings

- The encoder outputs frame-level hidden states with base dimension **2048**.
- Temporal pooling: compute mean μ and standard deviation σ over the time axis.
- Final utterance embedding:

$$z = [\mu\,;\,\sigma] \in \mathbb{R}^{4096}$$

  Concatenating mean and std doubles the dimension to **4096**, which is used for all downstream experiments.

Goal

Extract a fixed-length, TB-relevant audio representation from raw waveforms without any fine-tuning.

Classifier Evaluation Protocol

- Five classifiers are compared on the same embedding space:
  - Logistic Regression, MLP, SVM (RBF), Random Forest, Gradient Boosting.
- Each classifier is wrapped in a pipeline:

$$\text{StandardScaler} \rightarrow \text{PCA}(50) \rightarrow \text{Classifier}$$

- Cross-validation uses `StratifiedGroupKFold` with participant ID as group key, ensuring no patient appears in both train and validation folds simultaneously.
- Training is done on CODA (DB0); external evaluation is done on CIRDZ (DB1).
- Final reported score: mean AUROC across folds, then averaged across all classifiers.

Goal

Evaluate TB detection quality at each pipeline stage using a consistent, leakage-free protocol across multiple classifiers.

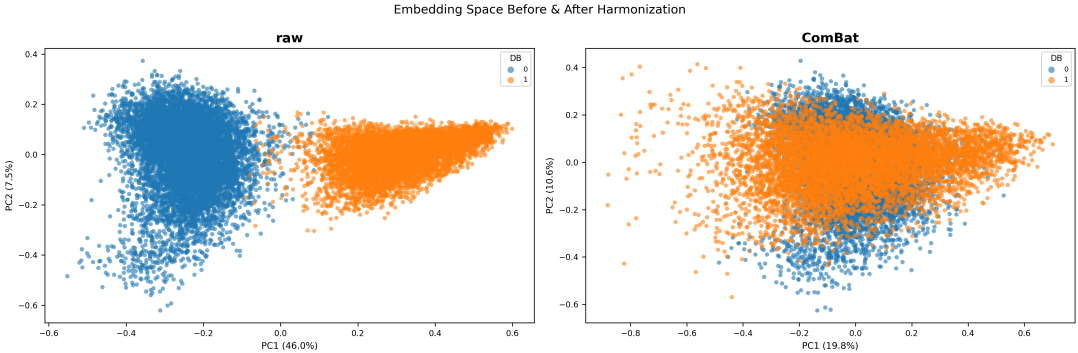
ComBat Harmonization

- ComBat is an empirical Bayes batch-effect correction method originally developed for genomics.
- Each database is treated as one batch. It estimates and removes:
  - additive batch shift $\hat{\gamma}$ (location),
  - multiplicative batch scale $\hat{\delta}$ (variance).
- Harmonization quality is evaluated with Ridge-based predictability of database labels in PCA space:
  - Raw: $R^2_{\text{db}} = 0.0198$, $R^2_{\text{dis}} = 0.0023$
  - ComBat: $R^2_{\text{db}} \approx 2.16 \times 10^{-8}$, $R^2_{\text{dis}} = 0.0017$
- After ComBat, database identity (CODA vs. CIRDZ) is no longer predictable from the embedding space, while disease information is partially preserved.

Purpose

Reduce dataset-specific (batch) variation in embeddings so that downstream classifiers focus on TB-related acoustic structure rather than database identity.

ComBat Result



Caption: PCA projections of embeddings before (left) and after (right) ComBat harmonization. Database-related variation is reduced after correction. External AUROC on CIRDZ improves from 0.4994 to 0.5119.

Outlier Detection Strategy

- Assumed contamination rate: **5%**.
- Two complementary scores are combined:
  - **Isolation Forest**: detects globally anomalous points in PCA-reduced (50-dim) embedding space.
  - **Per-database centroid distance**: flags samples whose distance from the database centroid exceeds $\sigma_{\text{threshold}} = 2.5$ standard deviations, normalized by Median Absolute Deviation.
- Weighted ensemble outlier score:

$$s = 0.55\, s_{\text{IF}} + 0.45\, s_{\text{centroid}}$$

- Samples in the top 5% of s are removed before the next stage.

Purpose

Remove samples whose embeddings are unlikely to represent clean cough signals, such as noisy recordings or non-cough acoustic content.

Outlier Removal Result

Caption: Left: detected outliers (×) overlaid on PCA projection. Right: retained inlier cloud after cleaning. Removed **836** samples (5.0%). Mean AUROC across classifiers increases to 0.6683 and stabilises the embedding distribution before TB subtype modeling.

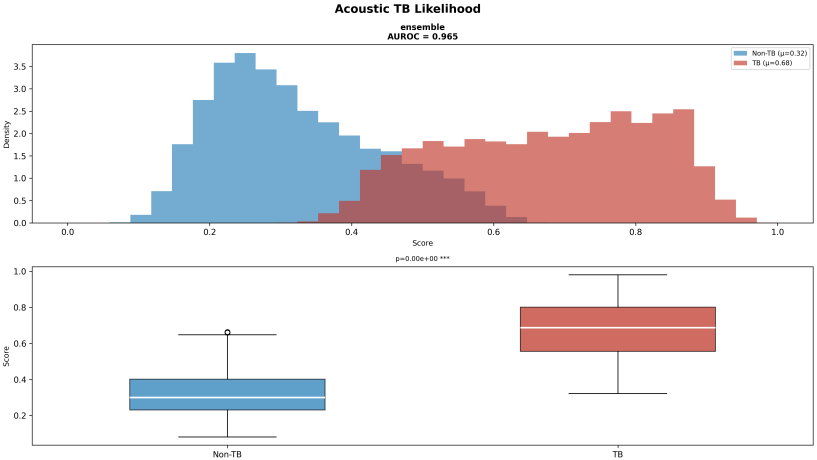
Acoustic TB Likelihood Modeling

- **kNN TB score** — makes no distributional assumption; judges each sample by its local neighborhood:
  - Compute mean distance to k TB neighbors (d_{TB}) and to k Non-TB neighbors (d_{NonTB}).
  - score $= d_{\text{NonTB}}/(d_{\text{TB}} + d_{\text{NonTB}})$. High score $\Rightarrow$ closer to TB cluster $\Rightarrow$ higher TB likelihood.
- **GMM posterior** — fits class-conditional density models to handle multi-modal acoustic subtypes:
  - Fit separate GMMs on TB and Non-TB embeddings (components selected by BIC).
  - Score $= P(\text{TB} \mid x)$ via log-sum-exp stable Bayesian posterior. High score $\Rightarrow$ higher TB likelihood.
- Final acoustic TB likelihood: simple average of kNN and GMM scores.

Goal

Assign each sample an unsupervised score reflecting how acoustically TB-like it is in the embedding space, to enable label refinement without additional annotations.

Acoustic TB Likelihood Result



Caption: Top: score distributions for TB and Non-TB. Bottom: box plots with Mann–Whitney U significance test. Ensemble mean scores: TB $= 0.677 \pm 0.146$, Non-TB $= 0.323 \pm 0.117$.

Relabeling Strategy

- Only original TB-labeled samples are relabeled; Non-TB samples are unchanged.
- Threshold defined as the median acoustic score among TB samples:

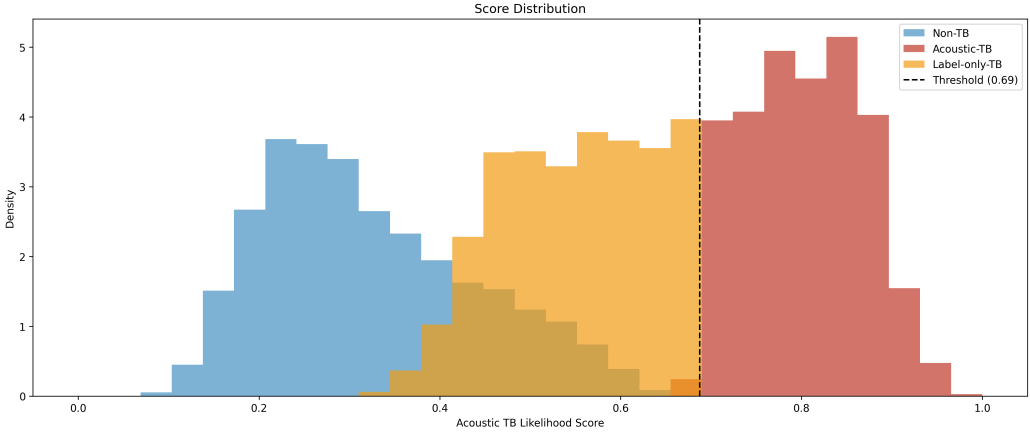
$$\tau = \mathrm{median}\big(\{s_i : y_i = \mathrm{TB}\}\big)$$

- Interpretation by position relative to τ:
  - score $\geq \tau \Rightarrow$ **Acoustic-TB**: embedding strongly matches the TB acoustic profile.
  - score $< \tau \Rightarrow$ **Label-only-TB**: annotated as TB but acoustically ambiguous; excluded from training.
- This is unsupervised label refinement, not class balancing.

Purpose

Refine noisy TB annotations by retaining only samples that are both annotated as TB *and* acoustically consistent with the TB embedding cluster.

Relabeling Result



Caption: Score distribution after relabeling. Left of the dashed threshold τ: Label-only-TB (acoustically ambiguous, excluded). Right of τ: Acoustic-TB (kept). Acoustic-TB: 1893 (50%), mean score 0.8030. Label-only-TB: 1893 (50%), mean score 0.5518.

Classification Performance (AUROC)

Stage	Mean AUROC (CV, CODA)	Mean AUROC (External, CIRDZ)
Baseline	0.6613	0.4994
After ComBat	0.6623	0.5119
After Outlier Removal	0.6683	0.5094
After Acoustic-TB Filtering	0.7993	0.7197

Summary: ComBat and outlier removal provide incremental gains. Acoustic TB relabeling is the dominant step, yielding a +13% gain in cross-validation AUROC and a +22% gain on external DB1 evaluation.